

Premiums to employment in larger establishments: evidence from Peru

Julie Anderson Schaffner *

*Center for Research on Economic Development and Policy Reform, Stanford University, Stanford,
CA 94305-6015, USA*

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Abstract

Examination of the employer size–wage effect for males in urban Peru suggests the following. Higher wages in larger establishments do not merely compensate for higher worker quality or inferior working conditions. Size differences in the use of rising wage–tenure profiles cannot provide a market-clearing explanation for the employer size–wage effect unless workers are extremely impatient. Thus, larger employers appear to pay premiums. Unionization explains none of the premiums, and minimum wage compliance explains at best a small fraction. Shirking-based efficiency wage explanations for premiums are implausible because Peruvian job security legislation greatly inhibits dismissals in larger establishments. Explanations meriting further research involve the use of high wages to reduce turnover and to improve productivity-enhancing morale. © 1998 Elsevier Science B.V.

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1. Introduction

In developing and developed economies alike, workers in larger establishments receive better wages and benefits than similar workers in smaller establish-

* Corresponding author. Tel.: +1-650-723-3978; fax: +1-650-723-8611; e-mail: julie@leland.stanford.edu.

ments.¹ The correlation between wages and size, known as the ‘employer size–wage effect,’ provokes research because it raises the possibility that wages above market-clearing levels render workers in larger establishments better off than identical workers in smaller establishments. Such violation of the Law of One Price would imply inefficiency in the allocation of labor and inequality of an especially troubling sort, in which wages differ across equally productive workers who differ only in luck or connections. Before concluding that larger employers pay premiums and labor markets are segmented, however, researchers must examine several potential explanations for the employer size–wage effect that are consistent with market clearing. Higher wages may, for example, merely compensate for higher worker quality or inferior working conditions.² If these explanations are inadequate, and larger employers appear to pay premiums, then researchers hoping to draw policy conclusions must press further, because premiums may arise for reasons with very different analytical implications. Labor unions or governments may compel larger employers to pay premiums, or larger employers facing more fundamental labor market imperfections may use high wages to increase worker effort, to reduce turnover, or to increase productivity in other ways suggested by the efficiency wage literature.

Determining which of the potential explanations for the employer size–wage effect are most important may be especially valuable for developing countries. Wage differences by establishment size are larger in developing countries than in the United States (see below), and there are many reasons for worrying that the larger wage differences reflect more severe labor market frictions. Government intervention in product, credit and labor markets is often more ambitious and less evenly enforced in developing countries; and information, risk and contracting problems are likely to be more severe in countries with weaker transport and communication systems, lower and more variable school quality, greater importance of uncertainties related to weather and illness, and weaker legal systems. Identifying frictions that cause larger employers to pay large premiums might help explain why large shares of developing country labor forces are employed in very small establishments or are self-employed,³ and might help to identify obstacles to economic development. Despite the potential importance to developing coun-

¹ Many studies have documented the size–wage association for the US, including Mellow (1982), Barron et al. (1987), Brown and Medoff (1989) and Kruse (1992). For evidence on West Germany, see Schmidt and Zimmermann (1991). Evidence to date for developing countries is somewhat sketchier, but see Little et al. (1987) on India and Malaysia, Aw and Batra (1994) on Taiwan, Fajana (1975) on Nigeria, Smith (1987) on Brazil, and Velenchik (1995) on Zimbabwe.

² As discussed in Section 3.2, larger employers’ greater use of long-term contracts with rising wage–tenure profiles may provide an additional market-clearing explanation for the employer size–wage effect. Although consistent with market clearing, this explanation raises the possibility that credit constraints differentiate workers in a way that is just as disturbing as labor market segmentation per se.

³ On the size distribution of employment in developing and developed countries, see Little et al. (1987). On cross-country patterns in rates of self-employment, see Schultz (1990).

tries, most research examining potential explanations for the employer size–wage effect pertains to the United States and Europe.

This paper examines the employer size–wage effect for urban males in Peru, using rich household survey data collected by the World Bank and the Peruvian government between July 1985 and June 1986. While no single developing country in any one year is typical of all other countries and years, Peru in 1985–1986 shared with many developing countries the extensive government intervention and weak physical and social infrastructure mentioned above, and was between extremes of bust and boom. Orthodox austerity, terms of trade shocks and bad weather produced a severe recession that bottomed in 1983, and the policies of the Garcia administration (which entered in July 1985) led to boom, hyperinflation and bust near the end of the decade. But 1985–1986 brought only declining inflation and mild growth (Pastor, 1992). In its first year, the Garcia administration explicitly aimed at redistribution toward the ‘informal’ sector, or producers outside the scope of government regulations who tend to operate at a small scale. A heterodox stabilization plan brought price controls, which were relevant only for formal producers, and special subsidized credit programs aimed at stimulating informal sector producers (Paus, 1991; pp. 416–417). As a result, smaller establishments may have been healthier and better able than at other times to pay high wages. Thus, if the employer size–wage effect fluctuates in magnitude, one might guess that it was at an ebb in 1985–1986, making its large size especially interesting.

The study, which examines a more complete list of potential explanations for the employer size–wage effect than most previous studies,⁴ yields important insights into the operation of labor markets in Peru and similar developing countries. It suggests the following. Higher wages in larger establishments cannot be explained by differences in worker quality or working conditions. Differences in the importance of specific human capital or of contracting problems leading to the payment of rising wage–tenure profiles may be important, but they cannot provide an explanation consistent with market clearing unless workers are extremely impatient. Thus, larger employers appear to pay premiums, which become large at a very low establishment size threshold (6 workers), which rise more slowly with size above that threshold, and which are larger for white-collar than blue-collar workers. Unionization explains none of the premiums, and compliance with minimum wage legislation explains at best a small fraction. Rather, premiums

⁴ Most studies for developed countries address only a subset of the explanations considered in the current paper. The exception is the study of Brown and Medoff (1989) for the United States. Although some papers have documented the employer size–wage effect for developing countries, few have attempted to explain it. An exception is the study of Velenchik (1995) for Zimbabwe. Although the studies differ greatly in country context, data sources and methodological details, results in the current study, Brown and Medoff (1989) and Velenchik (1995) lead to very similar conclusions about likely explanations for the employer size–wage effect.

arise out of more fundamental labor market imperfections, such as those highlighted in the literature on efficiency wages (Akerlof and Yellen, 1986). Among efficiency wage explanations, the most popular—the shirking model—may be ruled out. Job security legislation renders dismissals in larger establishments costly if not impossible, so it is implausible that larger employers use high wages combined with the threat of firing workers caught shirking to induce workers not to shirk. The most plausible efficiency wage variants involve larger employers' use of high wages to reduce turnover costs or to increase productivity-enhancing morale.

Section 2 documents the wage, benefits and job tenure patterns by establishment size that motivate the analysis. Section 3 examines explanations for the employer size–wage effect that are consistent with market clearing, some of which are consistent with spot labor market perfection, and some of which relate to larger employers' use of long-term contracts with rising wage–tenure profiles. Section 4 examines explanations for the payment of premiums. Section 5 suggests implications for future research.

2. Basic wage and benefits patterns

Empirical patterns motivating the analysis are derived from the Peruvian Living Standards Survey (PLSS) of 1985–1986 (see Grootaert and Arriagada, 1986 for a description). The tables pertain to 1132 males from the survey's representative sample of 5000 households, who were over age 14, lived in urban areas, were wage employees in private-sector establishments with at least two workers, reported positive wages, and answered several other questions. Appendix A describes the variables. Establishment size categories are derived from workers' reports of the number of workers at their place of employment. The categories are: micro (2 to 5 workers), small (6 to 20 workers), medium (21 to 200 workers) and large (over 200 workers).⁵

The focus of the paper is on explaining the large wage differences by establishment size seen in Table 1. Mean hourly cash wages more than double in the move from micro to large establishments, and mean hourly total wages (which include workers' reports of the cash value of in-kind payments) more than triple. Far from being driven by a few outliers, mean wages that rise with establishment size reflect increases at all quantiles of the wage distribution. Table 1 also shows that wage differences are part of a broader picture of more attractive employment

⁵ Firm size, which may have an independent effect on wages, is not measured independently. Brown and Medoff (1989) find independent effects, while Kruse (1992) finds both firm size and work group size insignificant after controlling establishment size.

Table 1
Descriptive statistics by establishment size^a

	All	Micro	Small	Medium	Large
Number of observations	1132	346	304	317	165
Percentage	100	311	27	28	15
Mean cash wage	4.63 (6.17)	2.92 (3.24)	4.25 (5.22)	5.72 (6.96)	6.85 (9.14)
Mean total wage	5.98 (13.29)	3.56 (3.86)	4.92 (5.71)	7.00 (8.73)	11.06 (30.64)
<i>Total wage quantiles</i>					
90th	11.30	6.29	9.46	14.50	15.45
75th	6.17	4.23	5.66	7.27	9.85
50th	3.62	2.52	3.32	4.35	6.13
25th	2.27	1.71	2.30	2.29	3.61
10th	1.59	1.01	1.69	2.03	2.18
<i>Percent with:^b</i>					
Paid holiday	51	20	48	73	80
Sick leave	48	17	42	71	80
Pension	45	13	38	68	81
Mean years tenure	6.45 (8.01)	5.33 (7.39)	5.60 (7.14)	6.78 (8.57)	9.31 (8.87)
<i>Years tenure quantiles</i>					
90th	20	18	17	20	23
75th	10	7	8	10	15
50th	3	2	2.71	3	6
25th	0.75	0.42	0.83	1	2
10th	0.17	0.17	0.25	0.25	0.33
Mean highest grade attained	9.40 (3.53)	8.74 (3.23)	9.06 (3.52)	9.83 (3.70)	10.62 (3.41)
Mean potential experience	18.67 (14.14)	17.01 (15.53)	17.85 (12.91)	19.90 (13.91)	21.28 (13.20)
<i>Percentage</i>					
Household head	53	44	51	56	67
Blue-collar	63	69	66	57	55

^aNumbers in parentheses are standard errors.

^bBased on 1120 observations.

Table 2
OLS log total wage regression coefficients^a

Sample	All	Blue-collar	White-collar
Number of observations	1132	711	421
Constant	−0.514 ** (0.095)	0.047 (0.132)	−0.558 * * (0.168)
School	0.111 * * (0.007)	0.055 * * (0.011)	0.115 * * (0.012)
Experience	0.034 * * (0.006)	0.029 * * (0.007)	0.030 * * (0.010)
Experience ² /100	−0.033 * * (0.011)	−0.034 * * (0.013)	−0.026 (0.017)
Household head	0.174 * * (0.052)	0.148 * * (0.061)	0.294 * * (0.084)
Small	0.249 * * (0.056)	0.172 * * (0.067)	0.397 * * (0.092)
Medium	0.412 * * (0.055)	0.293 * * (0.065)	0.603 * * (0.093)
Large	0.550 * * (0.067)	0.462 * * (0.077)	0.734 * * (0.112)
R ²	0.358	0.178	0.425
Adjusted R ²	0.354	0.170	0.416

* Significant at 10% level.

* * Significant at 5% level.

^aNumbers in parentheses are heteroskedasticity-consistent standard errors.

conditions in larger establishments. Workers in larger establishments are much more likely to be entitled to paid holiday, paid sick leave and pensions; and job tenure data suggest longer-lasting employment relationships in larger establishments.⁶

As in the US (Mellow, 1982), larger establishments employ workers who are on average more highly schooled, more experienced and more likely to be household heads; but regressions presented in Table 2 illustrate that large wage differences by size remain after controlling these differences in worker characteristics. The dependent variable in this and all other tables is the log of the total wage; results are similar if the log of the cash wage is substituted. Coefficients on establishment size indicators in column 1 imply that employees in small, medium and large establishments earn on average 25, 53 and 64% higher wages than employees in micro establishments, after controlling for education, experience and household headship. The size effects are stronger for white-collar workers (column

⁶ The tenure variable requires some discussion. Respondents were asked to describe their main job in the last 7 days, and were then asked how long they had been doing the specific activity they described. They were asked to report numbers of full years plus additional months. This variable is used as a job tenure variable in van der Gaag et al. (1989), but in fact there is some ambiguity as to whether this is a job tenure variable or an occupational tenure variable. To the extent workers have more scope for changing occupations within large establishments (for empirical evidence from the US, see Idson, 1989), differences in these tenures across establishment sizes are likely to understate the extent to which length of employment in the establishment rises with establishment size. On the other hand, the positive association of tenure and size could be overstated if larger establishments tend to hire workers who had more experience doing exactly the same activity in other firms.

3) than for blue-collar workers (column 2).⁷ The three size coefficients are jointly statistically significant, and the hypothesis that the three coefficients are identical is rejected, in all regressions in Table 2. Quantile regressions (not shown) indicate a shifting of the entire wage distribution as establishment size rises. Twenty-fifth, fiftieth and seventy-fifth quantile regressions are similar to the mean regressions, while size effects are somewhat larger in the lower tail of the wage distribution (the tenth quantile) than in the upper tail (the ninetieth quantile).

Peruvian establishment size–wage associations exceed those found in the United States. Regrouping the Peruvian data into employer size ranges of 2–10, 11–50, 51–100, and over 100 workers allows comparison with the May 1988 Current Population Survey data (on male urban workers) grouped into ranges of 1–9, 10–49, 50–99, and at least 100 workers. Fifty percent of the CPS sample, and 71% of the Peruvian sample, falls into the smallest two ranges. When dummies for the three largest ranges are included in log wage regressions similar to those in Table 2, the coefficients for the US are 0.143, 0.215 and 0.297, while the coefficients for Peru are 0.291, 0.340 and 0.425 (all strongly significant).⁸ The larger Peruvian coefficients primarily reflect a much wider wage gap between micro and small establishment wages.⁹ The most pronounced size effects in both countries are at the low end of the size spectrum.

Few closely comparable estimates exist for other developing countries, but available estimates suggest that the employer size–wage effect is large in other parts of the developing world as well. Male workers in Bombay, India, in establishments of over 500 employees, earn on average 42% more than employees of establishments with 10–99 workers, after controlling for workers' schooling, age and indicators of training, language and blue-collar status; and Malaysian workers in enterprises of 1–9 workers earn 32% less than those in larger enterprises, after controlling for similar worker traits (Little et al., 1987; pp. 256, 275). When Zimbabwean workers are divided into the size categories of Table 2, similar regressions produce even higher coefficients than those found in Peru (Velenchik, 1995).

Size differences in the incidence of nonwage benefits and in job tenures also remain large and significant in Peru after controlling for standard worker productivity proxies. Probit estimates imply that, for a representative individual with mean schooling and experience and who is a household head, the probability of

⁷ Blue- and white-collar wages are distinguished, because some of the explanations for the employer size–wage effect discussed below are of differential relevance for blue and white collar workers. The hypothesis that all coefficients are identical for blue- and white-collar workers is strongly rejected.

⁸ The CPS regression differs from the Peruvian sample in two ways. First, it does not exclude employees who work alone in their workplaces (a very small number in Peru). Second, the CPS wage is gross of taxes and social security payments, while the Peruvian wage is more nearly a net measure.

⁹ Differences between medium and small coefficients are 0.071 in the United States and 0.049 in Peru, while differences between large and medium coefficients are 0.082 and 0.085.

receiving a pension is 30, 57 and 65 percentage points higher in small, medium and large establishments relative to the probability in a micro establishment. Predicted probabilities of receiving paid holiday and sick leave vary in similar fashion across sizes, and the incidence of all benefits rises significantly with size in both blue- and white-collar samples. Maximum likelihood estimates of job tenure duration models based on the assumptions that the distribution of completed job tenures (i.e., tenures when jobs end) for jobs beginning in any period is of the Weibull form, and that this distribution and the rate of flow into new jobs has been constant over time, imply that blue-collar workers in small, medium and large establishments are 4, 17 and 31% less likely to quit or be fired in any period than workers in micro establishments (only the medium and large coefficients significantly differed from zero). The comparable figures for white-collar workers (all significantly different from zero), are 14, 23 and 55% (details available from the author).

3. Market-clearing explanations for the employer size–wage effect

3.1. *Explanations consistent with perfect spot markets*

Average wages may be higher in larger establishments even if equally productive workers receive the same wages everywhere, if workers in larger establishments are, on average, of higher productivity (after controlling for schooling, experience and household headship). Wages received by identical workers may also be higher in larger establishments without leaving the workers there better off, if the high wages merely compensate for inferior working conditions.¹⁰ Larger establishments' higher worker quality requirements or inferior working conditions may arise out of their more modern, capital-intensive techniques of production or more complex organizational forms. If the employer size–wage effect is explained by differences in worker quality or working conditions, then, not only does the employer size–wage effect provide no evidence of labor market segmentation; it provides no evidence of any departure from spot labor market perfection.

¹⁰ If workers differ in preferences toward working conditions, then one might expect workers more averse to bad working conditions to be sorted into establishments with better working conditions; but equilibrium might still require higher wages in establishments with worse working conditions. In equilibrium, workers with identical productivity and preferences are equally well off anywhere they are observed to work, but only workers with marginal preferences work under both good and bad working conditions, and most workers strictly prefer their own jobs to any other in the economy. If larger employers pay higher wages to compensate for inferior working conditions, then lower costs of capital, output quality premiums or some other advantage must allow them to coexist with smaller employers in equilibrium.

3.1.1. Worker quality selection and log wage regressions

If higher wages in larger establishments reflect higher average worker quality, then the inclusion of more complete controls for worker quality in regressions like those in Table 2 should cause the establishment size coefficients to become small and insignificant. Making use of the PLSS data, I can improve controls for worker heterogeneity in several ways. In addition to allowing the effect of years of schooling to differ across primary, secondary and post-secondary levels, and allowing discrete effects of obtaining university degrees or degrees from other post-secondary institutions,¹¹ I can introduce controls for whether the last school the individual attended was public or private (a crude proxy for quality of education),¹² indicators of whether the individual reported receipt of training on or off the job, and of whether he received certification in these sorts of training,¹³ as well as controls for the years of schooling of the individual's father¹⁴ and the individual's occupation.

Allowing the schooling effect to differ by level may be important, either because the productivity effects of schooling vary across levels of schooling, or because the average quality of schools or of students differ across levels. Similarly, diploma and training effects may capture either higher acquired skills or higher innate productivity of the individuals who acquire the diplomas or training. Either way, the controls are useful for producing size effect estimates purged of worker quality differences. Father's education may capture higher investments in their children's human capital by more educated parents. But because significant parental education effects may also indicate the importance of 'connections,' which are valuable only in markets in which some jobs pay premiums, their inclusion may lead to an overstatement of the importance of worker quality differences in explaining wages and an understatement of the apparent premiums. Similarly, occupational titles may capture variation in innate or acquired skills, but their inclusion could lead to an understatement of size premiums if organizational

¹¹ I also allowed diploma effects in the last years of primary and secondary school, but they were not significant, and allowing for them had no effect on the size coefficients.

¹² The percentage of workers reporting public school attendance falls with size for white-collar workers, from 88% in micro establishments to 65% in large, but are less systematic across sizes for blue-collar workers.

¹³ For both blue- and white-collar workers, the share of workers reporting off-the-job training (see definition in Appendix A) rises by over 10 percentage points from micro to small establishments, and varies little among small, medium and large establishments. The percentage reporting completion of on-the-job training rises from 1 to 14% for white-collar workers and from 1 to 6% for blue-collar workers. On-the-job training might produce general or specific skills. Percentages with certification in such types of training vary similarly.

¹⁴ For white-collar workers, average father's schooling rises monotonically with size, from 6.6 years in micro establishments to 9.4 years in large establishments. For blue-collar workers, the averages vary much less by size and exhibit no clear pattern. I also included mother's education, which varies similarly by size; but doing so made little difference.

requirements leading larger employers to pay premiums also lead them to use a different structure of job titles.

Table 3 summarizes the effects on size coefficients in blue- and white-collar log wage regressions of improving controls for worker heterogeneity. To facilitate comparison, columns 1 and 5 repeat the results reported in the last two columns of Table 2, suppressing all but the size coefficients. Columns 2 and 6 repeat the simple regressions on the smaller samples for which all worker quality controls are available, while columns 3 and 7 introduce all worker quality controls except the occupation code indicators. Experimentation with a wide range of interaction effects had little effect on the estimated size effects; thus I kept the table short by reporting a simpler specification. Finally, columns 4 and 8 add two-digit occupation controls.

For blue-collar workers, the controls introduced in column 3 have little effect on the size coefficients, although some controls have large and statistically significant wage effects. The largest marginal effect on size coefficients is associated with adding the technical degree indicator, which brings the largest size coefficient down by several percentage points. Introducing occupation controls has little effect on the largest size coefficient, but brings down the small and medium coefficients (and widens the gap between medium and large establishment wages) by several percentage points. For white-collar workers, the controls introduced in column 7 bring about larger reductions in size coefficients than they did for blue-collar workers. The strongest marginal effects are associated with father's education, which reduced all size coefficients, and the public school indicator and the training indicators, which tend to bring the large establishment coefficient down. Again, introducing occupation controls reduces small and medium size coefficients and increases the gap between medium and large wages. Overall, size coefficients remain large and significant, and larger for white-collar than for blue-collar workers.

Two additional methods of controlling for worker quality selection are possible with the PLSS data, although they prove somewhat less useful than is sometimes hoped. First, one can control for some dimensions of unobserved ability by estimating a model with family fixed effects on a subsample of brothers for whom common fathers may be identified. If the most important unobserved abilities are shaped by heredity or upbringing, and are closely correlated across brothers, and if the model is otherwise well specified, then family fixed effects methods should provide unbiased size coefficient estimates.¹⁵ Unfortunately, men can be identi-

¹⁵ If family components account for a larger fraction of variance in unobserved factors influencing establishment size but not wages directly (say, through family location) than of variance in the ability factors influencing both size and wages, then the fixed effects estimates will be more biased than OLS. Biases introduced by errors in variables or correlation between regressors and the error (after controlling for family effects) are augmented when differencing reduces the signal-to-noise ratio. See Griliches (1979).

Table 3
OLS log total wage regression coefficients^a

Sample	(1) Blue-collar	(2) Blue-collar ^b	(3) Blue-collar ^b	(4) Blue-collar ^b	(5) White-collar	(6) White-collar ^b	(7) White-collar ^b	(8) White-collar ^b
Number of observations	711	654	654	654	421	401	401	401
Small	0.172 ** (0.067)	0.182 ** (0.071)	0.176 ** (0.069)	0.130 * (0.069)	0.397 ** (0.092)	0.395 ** (0.095)	0.329 ** (0.096)	0.268 * (0.010)
Medium	0.293 ** (0.065)	0.288 ** (0.068)	0.277 ** (0.068)	0.233 * (0.069)	0.603 ** (0.093)	0.612 ** (0.095)	0.525 ** (0.089)	0.477 * (0.096)
Large	0.462 ** (0.077)	0.484 ** (0.081)	0.452 ** (0.080)	0.457 ** (0.084)	0.734 ** (0.112)	0.763 ** (0.117)	0.593 ** (0.109)	0.612 * (0.118)
Years			0.030 (0.032)	0.030 (0.032)			−0.066 (0.155)	0.307 * (0.143)
primary								
Years			0.048 ** (0.016)	0.059 ** (0.016)			0.064 ** (0.025)	0.029 (0.031)
secondary								
Years			−0.059 (0.039)	−0.055 (0.036)			0.072 ** (0.023)	0.063 * (0.025)
post-second								
University								
degree			0.925 * (0.517)	0.773 (0.809)			0.270 * (0.138)	0.260 * (0.138)
Technical								
degree			0.576 ** (0.226)	0.643 * (0.244)			0.132 (0.138)	0.248 * (0.148)
Public school			−0.165 * (0.109)	−0.064 (0.102)			−0.074 (0.089)	−0.069 (0.086)
Father's			0.021 ** (0.008)	0.017 * (0.008)			0.048 ** (0.010)	0.041 ** (0.009)
schooling								
Off-job			0.087 (0.079)	0.064 (0.079)			−0.202 * (0.114)	−0.179 (0.114)
training								
Off-job			0.080 (0.104)	0.044 (0.096)			0.333 ** (0.120)	0.284 * (0.125)
certification								
On-job			−0.100 (0.340)	0.010 (0.488)			0.499 (0.316)	0.218 (0.216)
training								
On-job			0.055 (0.541)	−0.093 (0.686)			0.029 (0.363)	0.128 (0.225)
certification								
Occupation	No	No	No	Yes	No	No	No	Yes
Occupation								
dummies ^c								
R ²	0.178	0.159	0.190	0.292	0.425	0.420	0.491	0.606
Adjusted R ²	0.170	0.149	0.169	0.210	0.416	0.410	0.469	0.520

* Significant at 10% level.

** Significant at 5% level.

^a Regressions also include years of schooling, potential labor market experience and an indicator of household headship. Numbers in parentheses are heteroskedasticity-consistent standard errors.^b Sample excludes workers in mining and electricity, and workers for whom any worker quality controls are missing.^c Occupation dummies control for two-digit occupation categories.

fied as brothers only if their father is present in the surveyed household, thus the sample is small (35 brother pairs, 3 groups of 3 and 2 groups of 4) and differs systematically from the population. Analysis of the small sample suggests that worker selection across establishment sizes probably is not random,¹⁶ but fixed effects estimates are very imprecise. The point estimates are consistent with substantial unexplained size effects, but standard errors are sufficiently large as to admit both large and small size effects. Fixed effects estimates of the size coefficients (standard errors) are 0.256 (0.192), 0.203 (0.170) and 0.499 (0.329).¹⁷

Second, one may control for endogenous selection of workers across establishment size categories using techniques in the spirit of popular two-stage technique of Heckman (1979) (although more complicated here because selection is across four subsamples rather than two). In principle, these methods account for endogenous selection along dimensions more general than those captured by the inclusion of worker quality proxies or the treatment of family fixed effects, but only under strong assumptions on the statistical distributions of the unobserved qualities. Because results can be misleading if these assumptions are incorrect (Goldberger, 1983), I implemented these methods under two equally reasonable sets of assumptions. The first assumed that selection across establishment sizes had the structure of an ordered probit model, and that the log wage regression error's expected value was linear in the expected value of the error underlying the probit (as in Idson and Feaster, 1990). The second assumed that the selection model was multinomial logit and the log wage regression error was joint normally distributed with an appropriate transformation of the selection model errors (following Lee, 1983). The results (not shown) cast great doubt on the usefulness of such methods. Standard errors are large, and point estimates of the size coefficients differ greatly across statistical assumptions and are sensitive to changes in the list of variables included in the first-stage selection model.

Despite the weakness of the fixed effect and endogenous sample selection correction results, it seems reasonable to conclude that worker quality selection is unlikely to explain the employer size–wage effect in Peru. If differences in worker quality explain the wage differences by size remaining in columns 4 and 8 of Table 3, then there must exist large differences across workers in exogenous qualities that are not only unobserved by the researcher and observed by employ-

¹⁶ A Breusch–Pagan Lagrange multiplier test rejects OLS in favor of a random effects specification ($\chi^2(1)$ statistic value 6.76) and a Hausman test rejects random effects in favor of fixed effects ($\chi^2(6)$ statistic value 14.95) in regressions including schooling, experience, experience squared, and small, medium and large size indicators on this subsample.

¹⁷ Not having panel data, I could not attempt methods involving individual fixed effects over time. Even if such methods were possible, they would be unlikely to resolve the worker quality selection question. Applications of panel data methods to US data have led to conclusions that selection does not explain the size–wage effect (Brown and Medoff, 1989; Gerlach and Schmidt, 1990), and that it does (Evans and Leighton, 1989).

ers, but also are poorly correlated with all of the worker quality controls already included in the regressions. I find such qualities difficult to imagine.

3.1.2. *Working conditions and log wage regressions*

If higher wages in larger establishments compensate for inferior working conditions, then the inclusion of better working condition controls in regressions like those in Table 2 should cause size effects to become small and insignificant. The data suggest a variety of systematic differences in working conditions across establishment sizes, for which better controls may be introduced. Larger establishments are more likely to be located in Lima, where the cost of living may be higher in a way not captured in the deflator used for constructing the real wage variable, and where costs of transportation to work may be higher.¹⁸ Average usual weekly hours are higher in Larger establishments.¹⁹ The lower incidence of workers who report that their boss is a relative may reflect a less ‘familial’ work environment in larger establishments.²⁰ Larger employers, who must recruit workers over wider geographical areas, may have to pay higher wages to compensate for higher commute costs, as suggested by higher average commute times and a higher proportion of workers taking motorized transportation to work in larger establishments.²¹ Blue-collar workers in medium and large establishments are somewhat less likely to report that they are receiving training than blue-collar workers in micro and small establishments, suggesting that larger employers may pay more to compensate for lower provision of general on-the-job training.²² Finally, the mix of establishments across industries, which are likely to differ in working conditions, differs by establishment size.²³

I introduced controls for these differences in a variety of ways, attempting to control for the endogeneity of usual weekly hours and of commute mode and

¹⁸ The percentages of micro, small, medium and large establishments located in Lima are 53, 60, 68 and 79, respectively.

¹⁹ Mean usual weekly hours (standard deviations) are 48.1 (17.8), 49.3 (16.6), 48.5 (14.1) and 50.3 (13.7) in micro, small, medium and large establishments, respectively. Median hours are 48, the legal work week, in all establishments. The incidences of part-time work (defined as usual weekly hours less than 35) are 17, 13, 9 and 4%.

²⁰ The percentages of workers with a relative for a boss are 26.7, 9.9, 6.3, and 1.2 in micro, small, medium and large establishments, respectively.

²¹ Median one-way commute times are 20, 25, 30 and 30 min in micro, small, medium and large establishments, respectively. The percentages of workers walking to work are 46, 32, 21 and 8, while the percentages of workers taking public or private motor transport are 51, 68, 76 and 90.

²² The percentages reporting that they are currently in training programs for blue-collar (white-collar) workers are 7.5 (1.7), 8.8 (2.5), 5.8 (3.3) and 6.8 (5.6) in micro, small, medium and large establishments, respectively.

²³ Larger establishments are more heavily involved in manufacturing, and less involved in wholesale and retail trade, construction and services.

time,²⁴ introducing both one and two-digit industry controls, and in some cases splitting the sample to look within groups for which working conditions are more homogeneous. But the simplest results are representative and are presented in Table 4. Columns 1 and 5 reproduce the basic regression results for blue- and white-collar workers found in the last two columns of Table 2. Columns 2 and 6 repeat the basic regressions, but change the sample in two ways. First, observations for which working condition controls are missing are excluded, with little impact on size coefficients. Second, workers in mining and electricity are excluded; examination of mean wages by size and industry revealed that these sectors were small, very high wage sectors in which the great majority of establishments are large, suggesting that they are best dropped when attempting careful study of intra-industry size effects. Eliminating them reduces the largest size coefficient by several percentage points, and renders them insignificantly different from the medium coefficients for both blue- and white-collar workers.

Columns 3 and 7 introduce the working condition controls themselves, while columns 4 and 6 include both working condition controls and the controls for worker heterogeneity discussed in Section 3.1.1. The reductions in the largest size coefficients, somewhat larger for white-collar than blue-collar workers, are produced primarily by the introduction of controls for location in Lima and for the workers' use of motor transport to work. The marginal effects of introducing several of the controls is to increase the size coefficients. The medium and large size coefficients become significantly different from each other for blue-collar workers. The signs on some of the coefficients are the opposite of what would be needed for the related working conditions to explain the size–wage association. For example, longer hours are associated with significantly lower hourly wages, possibly reflecting the importance of binding hours rations. In sum, small and medium size coefficients are robust to the inclusion of working condition controls, remaining large, significant and larger for white collar than blue-collar workers. The largest size coefficients are less robust, becoming insignificantly different from medium coefficients for white-collar workers.

Although the list of working conditions included in the regressions is far from complete, it seems reasonable to conclude that working condition differences are unlikely to explain the employer size–wage effect in Peru, except possibly at the upper end of the establishment size spectrum. If variation along some dimension of working conditions is to explain wage patterns, then conditions along that dimension must differ more between micro and small establishments than between small, medium and large establishments, should have greater variation in Peru than in the US, and of course should be 'worse' in the larger establishments. Some working conditions satisfying those requirements certainly remain uncontrolled in

²⁴ Household per capita nonlabor income, numbers of adult males and females in the household, and numbers of the individual's children aged 14 and under were used as instruments. Unfortunately, the instruments are weak.

Table 4
OLS log total wage regression coefficients^a

Sample	(1) Blue-collar	(2) Blue-collar ^b	(3) Blue-collar ^b	(4) Blue-collar ^c	(5) White-collar	(6) White-collar ^b	(7) White-collar ^b	(8) White-collar ^c
Number of observations	711	680	680	625	421	397	397	379
Small	0.172 * (0.067)	0.167 * * (0.069)	0.165 * * (0.064)	0.140 * * (0.066)	0.397 * * (0.092)	0.381 * * (0.092)	0.425 * * (0.093)	0.331 * * (0.102)
Medium	0.293 * * (0.065)	0.283 * * (0.068)	0.272 * * (0.066)	0.214 * * (0.067)	0.503 * * (0.093)	0.609 * * (0.094)	0.587 * * (0.101)	0.531 * * (0.100)
Large	0.462 * * (0.077)	0.424 * * (0.079)	0.391 * * (0.088)	0.420 * * (0.088)	0.734 * * (0.112)	0.662 * * (0.118)	0.542 * * (0.137)	0.539 * * (0.138)
Industry dummies ^d	No	No	Yes	Yes	No	No	Yes	Yes
Lima			0.172 * * (0.055)	0.178 * * (0.057)			0.186 * * (0.079)	0.127 (0.089)
Usual weekly hours			−0.014 * * (0.002)	−0.013 * * (0.002)			−0.009 * * (0.002)	−0.010 * * (0.002)
Boss is relative			0.066 (0.069)	0.089 (0.071)			0.171 (0.116)	0.152 (0.118)
Commute min/10			0.008 (0.021)	0.011 (0.021)			−0.047 (0.043)	−0.005 (0.009)
Motor transport			0.257 * * (0.077)	0.236 * * (0.080)			0.220 * (0.127)	0.129 (0.131)
Commute min/10 * motor transport			−0.008 (0.024)	−0.010 (0.024)			0.023 (0.048)	0.002 (0.006)
Training now	No	No	−0.052 (0.181)	−0.034 (0.194)	No	No	−0.003 (0.104)	−0.011 (0.113)
Worker quality controls	No	No	No	Yes	No	No	No	Yes
R ²	0.178	0.171	0.351	0.428	0.425	0.435	0.507	0.662
Adjusted R ²	0.170	0.163	0.313	0.324	0.416	0.425	0.461	0.546

* Significant at 10% level.

* * Significant at 5% level.

^a Regressions also include years of schooling, potential labor market experience and its square, and an indicator of household headship. Numbers in parentheses are heteroskedasticity-consistent standard errors.

^b Sample excludes workers for whom any working condition control is missing.

^c Sample excludes workers in mining and electricity, and workers for whom any worker quality or working condition control is missing.

^d Industry dummies control for two-digit industry categories.

the regressions of Table 4.²⁵ For example, micro employers (who work side by side with their employees) may provide their workers with informal general training of a sort that is not picked up by workers' reports of having participated in on-the-job training programs, and this training may be especially valuable in countries where workers have less and lower quality formal schooling, and where family histories in nonagricultural production are shorter. On the other hand, casual observation in developing countries suggests that the prevalence of some good working conditions—such as having a completely enclosed workplace, electricity and running water—is higher in small, medium and large establishments than in micro establishments. Controlling for such conditions might be expected to increase apparent premiums in larger establishments.

3.1.3. *Worker quality, working conditions and job tenure patterns*

As mentioned in Part II, job tenures are higher in larger establishments even after controlling for differences in basic worker characteristics. It may also be shown that the higher tenures in larger establishments are closely associated with the payment of higher wages; when the wage is included on the right hand side of the duration models, it is found to be significantly negatively associated with job exit rates, and its inclusion reduces the absolute magnitude of the size effects, although it does not cause them to disappear (details available from the author). Whether larger employers make greater use of long-term contracts with rising wage–tenure profiles (for reasons discussed in the Section 3.2) or are more likely to pay premium wages (for reasons discussed in Section 4), these patterns are easily explained, because both rising wage profiles and premium wages should render high-wage workers reluctant to quit and more likely to be observed at high tenures.²⁶

These patterns are harder to explain if higher wages in larger establishments merely reflect higher worker quality or inferior working conditions. Worker quality selection could explain the tenure patterns only if the quality selected by larger employers is 'low quit propensity,' or something closely correlated with it. But to explain wage and tenure patterns, individuals' inherent quit propensities would have to differ greatly, and employers would have to observe good proxies for those quit propensities and differentiate pay on the basis of them. Simple compensating differential explanations for the employer size–wage effect cannot explain the tenure patterns. If wage differences reflect only differences in working

²⁵ Some additional working conditions, such as shift work, Sunday work, travel requirements, physical strain, hazards and pleasantness of environment and of interpersonal relationships, have been examined for the US and Germany by Brown and Medoff (1989), Kruse (1992) and Schmidt and Zimmermann (1991), who find that they fail to explain the size–wage effect.

²⁶ It is possible that tenure differences by establishment size (not controlling for wages) are explained by something unrelated to wage determination, such as lower dismissal rates in larger establishments, or faster job creation in smaller establishments in the years just prior to the survey. But under such circumstances, it is difficult to explain the effects of introducing the wage into the duration analysis.

conditions for which they perfectly compensate, then workers should be equally well off everywhere, and there would be no reason for quit rates to differ by establishment size or by wage level.²⁷ Thus, tenure patterns provide indirect support for the conclusion that perfect spot market explanations for the employer size–wage effect are likely to be incomplete.

3.2. Explanations involving long-term contracts with rising wage–tenure profiles

For several reasons, larger employers might find it especially important to establish long-term employment arrangements under which wages rise with tenure. Their more sophisticated technologies and larger organizations may lead larger employers to place greater value on specific on-the-job training, which produces skills valuable only in the establishment of the employer who provides the training (Becker, 1962). Neither workers who fear dismissal nor employers who fear resignations will bear the full cost of such training; but the parties may have incentives to share the costs of such training under long-term contracts allowing them both to share in the benefits of the training. Such contracts generally involve wages that are lower than workers' alternative wages during training (at early tenures) and are higher than the alternatives at higher tenures. Rising wage–tenure profiles may also provide workers with incentive not to shirk (Lazear, 1981),²⁸ and may aid employers in attracting workers with unobservably lower propensities to quit (Salop and Salop, 1976). If wages in larger establishments start low but rise with tenure, then wages averaged over workers at all tenures may appear higher in larger establishments, even if markets clear (in the sense that workers are indifferent between taking jobs in larger and smaller establishments), because workers with rising wage–tenure profiles have lower resignation rates and tend to be observed at higher tenures when wages are high, and because discounting requires higher tenure wages to exceed alternative wages by more than lower tenure wages falling short of alternative wages.

In Table 5, I examine the possibility that higher wages in larger establishments for high-tenure workers are compensated by lower wages at early tenures. Because incentives to invest in specific training may be higher at lower tenures, I allow for

²⁷ As pointed out by a referee, less simplistic compensating differential stories might explain lower resignation rates in larger establishments and where wages are higher, although only in the presence of some contracting difficulties. As suggested by Duncan and Stafford (1980), if working conditions are public goods within establishments, if employers cannot differentiate wages across workers with different preferences toward those working conditions, and if there are costs to searching out workers with homogeneous preferences that rise with establishment size, then larger establishments may choose to search less and to pay wages high enough to retain workers with more heterogeneous preferences. A higher share of workers in larger establishments would be inframarginal, and resignation rates would be correlated with both size and wages.

²⁸ As discussed in Section 4, stringent job security legislation binding at least on larger employers renders dismissal threats incredible. Thus, the Lazear argument, like the shirking-based efficiency wage argument discussed in Section 4, is unlikely to be relevant for Peru.

Table 5
OLS log total wage regression coefficients¹

Sample	(1) Blue-collar	(2) Blue-collar	(3) White-collar	(4) White-collar
Number of observations	711	711	421	421
Small	0.249 * * (0.098)	0.304 * (0.184)	0.242 * (0.143)	0.011 (0.275)
Medium	0.288 * * (0.090)	0.045 (0.154)	0.456 * * (0.147)	0.034 (0.055)
Large	0.369 * * (0.111)	0.143 (0.163)	0.502 * * (0.179)	−0.001 (0.019)
<i>Micro tenure variables</i>				
Tenure	0.021 (0.016)		−0.007 (0.037)	
Tenure ² /10	−0.008 (0.005)		0.006 (0.018)	
Tenure; $t \leq 1$		−0.107 (0.172)		−0.067 (0.294)
Tenure; $1 < t \leq 5$		0.082 * * (0.040)		0.034 (0.055)
Tenure; $t > 5$		−0.013 (0.009)		−0.001 (0.019)
<i>Small tenure variables</i>				
Tenure	−0.007 (0.020)		0.064 * * (0.025)	
Tenure ² /10	0.002 (0.008)		−0.023 * * (0.008)	
Tenure; $t \leq 1$		−0.199 (0.178)		0.351 (0.215)
Tenure; $1 < t \leq 5$		0.048 (0.037)		0.061 (0.046)
Tenure; $t > 5$		−0.009 (0.011)		−0.011 (0.015)
<i>Medium tenure variables</i>				
Tenure	0.013 (0.011)		0.035 * (0.021)	
Tenure ² /10	−0.003 (0.004)		−0.005 (0.006)	
Tenure; $t \leq 1$		0.273 * (0.156)		0.497 * (0.260)
Tenure; $1 < t \leq 5$		0.040 (0.037)		0.030 (0.050)
Tenure; $t > 5$		−0.006 (0.007)		0.012 (0.009)
<i>Large tenure variables</i>				
Tenure	0.036 * (0.021)		0.076 * * (0.029)	
Tenure ² /10	−0.011 * (0.006)		−0.027 * * (0.011)	
Tenure; $t \leq 1$		0.187 (0.206)		0.630 * * (0.313)
Tenure; $1 < t \leq 5$		0.103 * * (0.048)		0.088 (0.073)
Tenure; $t > 5$		−0.012 (0.008)		−0.016 (0.015)
R^2	0.185	0.202	0.451	0.460
Adjusted R^2	0.168	0.180	0.431	0.434

* Significant at 10% level.

** Significant at 5% level.

¹Regressions also include schooling, potential labor market experience and its square and an indicator of household headship. Numbers in parentheses are heteroskedasticity-consistent standard errors.

nonlinearities in the tenure effects. Columns 1 and 2 refer to blue-collar workers, while columns 3 and 4 refer to white-collar workers. In columns 1 and 3, the tenure effects are quadratic. Experimentation with more flexible treatment of tenure effects involving piecewise linear splines over one- and 5-year tenure

ranges, although demanding the limited data and producing some erratic results, led me to conclude both that the estimates of size effects at zero tenure (the coefficients on the size indicators) are sensitive to functional form choice and that, at least for white-collar workers, tenure effects are especially strong in the first year on the job. To give the reader some of this in a parsimonious way, in columns 2 and 4, I present estimates allowing for three-part tenure splines (but not allowing for discontinuities between segments), in which piecewise linear wage–tenure relations may differ across tenure ranges of up to one year, from one through 5 years, and over 5 years.

The results indicate more important tenure effects in larger establishments, more so for white-collar than for blue-collar workers. With quadratic tenure effects, size effects at zero tenure (estimated by coefficients on the size indicators themselves) for blue-collar workers remain strong and significant between micro and small establishments, although they rise only slightly (and not statistically significantly) in medium and large establishments, while for white-collar workers zero-tenure wages continue to rise significantly with size. With three-part splines, tenure effects appear especially strong in the first year of tenure and there is little evidence that wages rise with size at zero tenure. For blue-collar workers, initial wages rise from micro to small establishments and then decline with size, while for white-collar workers, they become virtually flat across sizes. Which specification is preferable is debatable; the spline is more flexible in some ways and is less sensitive to outliers at high tenures, but its artificial structure may exaggerate the relative importance of first-year effects. Taking the two specifications together, one concludes that tenure probably matters, especially in large establishments and especially for white-collar workers, but that much of the size effect ‘hits early,’ either in wages at zero tenure, or within the first year of employment.

These results are consistent with market-clearing explanations for the employer size–wage effect involving rising wage–tenure profiles, but only if workers are credit-constrained and highly impatient. Although one could not reject the hypothesis that wages at zero tenure decline with establishment size (except when comparing blue-collar initial wages in micro and small establishments), one would reject the hypothesis that they decline by more than, say, 30%. Workers must be highly impatient if such reductions in initial wages leave them indifferent between jobs with and without training, in the face of subsequent training premiums of 60 or 80% that begin within a year and last throughout the job. Doubting that workers are that impatient, I am inclined to conclude that wage–tenure profiles in larger establishments contain premiums for reasons such as those discussed in Section 4.

Note, however, that distinguishing between the market-clearing explanations of this section and the non-market clearing explanations of Section 4 may be less important than distinguishing all of these from the explanations consistent with perfect markets (involving worker quality and working conditions). Although explanations involving rising wage–tenure profiles are consistent with market clearing, they nonetheless raise the possibilities of inefficiency and of troubling

inequities. Long-term contracting may not fully circumvent the underlying imperfections giving rise to the contracts. For example, the inability to contract completely over future random fluctuations in the worker's productivity inside and outside the current place of employment may lead to inefficient quits, dismissals and specific training investment decisions, even when contract parameters are chosen to minimize the inefficiency (Hashimoto, 1981). Furthermore, when workers must choose between jobs with upward-sloping and flat wage–tenure profiles, then credit constraints may lead to a phenomenon just as disturbing as labor market segmentation per se. In segmented labor markets, differences in luck or connections lead to differences in lifetime earnings among otherwise identical workers. In a world with rising wage–tenure profiles and credit constraints, differences in the wealth or credit constraints early in life may lead to differences in lifetime earnings for otherwise identical workers.²⁹

The apparent importance of tenure in wage determination might lead one to concentrate attention on explanations for premiums that also explain why larger employers would want to develop long-term contracts with rising wage–tenure profiles. Two caveats on the tenure results suggest, however, that explanations for premiums that could arise even in spot labor markets (such as those involving unionization and minimum wage compliance) also merit attention. First, the results of Table 5 do not allow one to reject the possibility that much of the size premium is enjoyed even by workers in new jobs. Second, cross-section correlations between wages and tenure may arise even if wages do not rise with tenure on individual jobs; workers receiving premiums that render them reluctant to quit may tend to be observed at higher tenures, causing average wages to be higher among higher tenure workers.

4. Explanations for larger employers' payment of premiums

4.1. Unionization and minimum wage compliance

In looking for reasons why larger employers pay premiums, it is natural to look first to labor unions and to incompletely enforced government wage regulations.³⁰ Rates of unionization and of (a proxy for) compliance with minimum wage

²⁹ An additional reason for not making too great a distinction between market-clearing and nonmarket-clearing explanations for the employer size–wage effect arises out of the observation that specific capital models become turnover-based efficiency wage models of labor market segmentation with only minor modifications, which acknowledge constraints preventing initial wages from being driven low enough to leave workers indifferent between jobs with and without upward-sloping wage–tenure profiles (Salop, 1979).

³⁰ As with compensating differential explanations, unionization and minimum wage compliance explanations of the employer size–wage effect are consistent with equilibrium in product markets only if establishments paying high wages enjoy advantages in other markets.

legislation rise with establishment size in Peru, suggesting that such explanations merit consideration, although the indicators and their potential importance given Peruvian circumstances require some discussion.

The unionization indicator derives from workers' reports of whether their establishments had a 'labor union or other workers' organization'. The percentages of workers whose workplaces had such organizations are 3, 8, 34 and 70 in micro, small, medium and large establishments, respectively. Unfortunately, the meaning of this indicator is likely to vary with establishment size. To become unions in Peru, workers' organizations must have at least 20 members and must be organized by firm and separately for blue- and white-collar workers (Gonzales, 1980, p. 225). Within-enterprise worker groups with between five and 20 members can elect a representative to negotiate with the employer, but are not given official union status. Thus, the few workers in micro and small establishments coded as 'unionized' are reporting a workplace organization without official union status, are in establishments that belong to multi-establishment firms with official unions, or have answered incorrectly.

Differential unionization might underlie the employer size–wage effect in several ways, with different implications for empirical attempts to discern its importance. If workers in unionized establishments command uniform premiums everywhere, and if unions underlie the employer size–wage effect merely because a larger share of large establishments is unionized, then the unionization indicator should have a positive effect in log wage regressions, and accounting for its effect should render size effects insignificant among both unionized and nonunionized establishments. If nonunion establishments facing greater threats of unionization pay higher wages to stave off unionization, and if the higher incidence of unionization in larger establishments indicates stronger unionization threats, then wages should rise with size among nonunion (but not union) establishments, and regression coefficients on unionization indicators should decline with size.³¹ If, however, the unions' ability to increase their own wages rise with establishment size, then wages might rise with size among both union and nonunion establishments, and coefficients on unionization indicators could be small for all establishment sizes. Thus, in exploring the role of unions in wage determination, one might want to allow union effects in wage regressions to differ by establishment size, but must be cautious in interpreting the results.

A closer look at the labor movement in Peru reduces the plausibility of differential unionization explanations for the employer size–wage effect, except possibly through a mechanism that is hard to distinguish from the wage regulation explanation discussed below. The focus of Peruvian union activity in the early and mid-1980s was on influencing national labor policy (minimum wages and job security regulations) rather than on plant-level activities of the sort that would

³¹ For a formal union threat model, see Dickens (1986).

create wage differences between unionized (or threatened) firms and others; and even in their priority activities they were weak. According to Bollinger (1987), the first half of the 1980s witnessed a decline in the number and length of strikes, a redirection of energy among veteran union members from union activities to finding supplementary employment and starting small businesses, and a redirection of union efforts toward influencing national policy and organizing national strikes, which often were dismal failures (pp. 624–625). Although President Garcia had courted labor unions during his campaign, he gave them little of what they wanted (Pastor, 1992; p. 126).

The proxy for whether the employer is complying with wage regulations is drawn from workers' reports of whether they are entitled to the benefits of the Instituto Peruano de Seguridad Social (IPSS). The incidence of IPSS is 19, 44, 75 and 85% in micro, small, medium and large establishments, respectively. Just as the labor law requires employers to pay at least the minimum wage, it requires that they register with the Ministry of Labor and pay a 5% wage tax in order to provide workers with IPSS health care, disability insurance and retirement benefits. Filing papers to comply with one regulation renders an establishment more likely to be monitored with regard to compliance with other regulations; thus, provision of IPSS benefits and compliance with minimum wage should be correlated. Consistent with this, the descriptive literature on formal and informal sectors in Peru treats compliance with labor standards as a package deal. The IPSS indicator may, however, be a better proxy for compliance with wage regulations in larger establishments; if the Ministry of Labor is more concerned with labor practices at larger establishments, then it is more likely to monitor practices for large establishments that sign up for IPSS than for smaller establishments that sign up.

Again one should consider several ways in which differential compliance with minimum wage law might help explain the employer size–wage effect. If establishments that are highly visible or place great value on obtaining government licenses must pay all workers at least the minimum wage, while all other establishments pay lower market-clearing wages, and if the incidence of compliance is higher among larger establishments, then an indicator of minimum wage compliance should have a positive effect in log wage regressions and its inclusion should cause size effects to become small and insignificant.³² The effect of compliance on wages could be large in Peru, where the legal minimum wage is quite high.³³ One might expect compliance primarily to increase the wages of

³² The coefficients might understate minimum wage effects on wages if IPSS confers valuable benefits for which workers are willing to take wage cuts.

³³ Over the 12 months during which the PLSS data were collected, the legal minimum monthly wage was 65 to 85% of the median hourly wage, assuming 48 h of work a week (the legal work week). The percentages of blue- (white-) collar workers receiving nominal wages below the legal minimum in micro, small, medium and large establishments are 52.3 (43.0), 34.2 (13.7), 20.6 (6.6) and 15.6 (4.0), respectively.

low-wage workers, although sociological and incentive considerations might lead complying employers to pay premium wages to some higher-wage employees as well. Thus, one might expect the compliance indicators to have larger coefficients in lower quantile regressions than in upper quantile regressions.

Threat effects might again complicate the story. If one workers' complaint to the Ministry of Labor is enough to get a non-compliant firm in trouble, and if larger employers find it more difficult to screen workers or to influence workers' propensity to file complaints with the Ministry of Labor, then larger non-IPSS establishments might pay higher wages to stave off complaints. Under these circumstances, one might expect to see size effects among non-IPSS (but not IPSS) establishments, and IPSS effects that decline with establishment size.

On the other hand, the endogeneity of compliance decisions provides good reason to think that IPSS effects will overstate the role of minimum wage legislation in explaining the employer size–wage effect. Employers choosing to pay high wages for reasons such as those discussed in Section 4.2 will find compliance with minimum wage and other labor legislation less costly, and are more likely to comply. Thus, IPSS may spuriously pick up the effect of conditions that would lead to the payment of high wages even in the absence of the legislation.

The effects of introducing unionization and IPSS indicators in log wage regressions for blue- and white-collar workers are shown in Table 6. Differentiating between blue- and white-collar workers might be important, because Peruvian unions are organized separately for blue- and white-collar workers, and because Peruvian labor legislation historically differentiated blue- and white-collar workers. Columns 1 and 6 repeat the size coefficient results from the basic log wage regressions. When the indicators are entered only linearly (in regressions not shown), they are insignificant and have little effect on the size coefficients. Columns 2 and 7 introduce unionization and IPSS differentiated by establishment size.

For blue-collar workers (column 2) the union coefficients are erratic and not jointly significant, but the IPSS effects are jointly significant and are substantially larger in medium and large establishments. Allowing for the IPSS effects reduces the medium and large size coefficients, rendering the small, medium and large coefficients insignificantly different from each other. The small coefficient and the three size coefficients jointly continue to be significantly positive. This qualitative picture is largely maintained when worker quality and working condition controls are added (in column 3), except that the IPSS effect in large establishments falls, suggesting that some of what appeared in column 2 as an IPSS effect may have been working condition differences. Introducing tenure effects brings down IPSS effect estimates only a little (not shown), but reduces them to insignificance as standard errors increase. For white-collar workers, neither the unionization nor the IPSS effects are jointly significant (column 7). This is also the case when worker quality and working condition controls are included (not shown). When three-part

Table 6
Log total wage regression coefficients^a

Sample	(1) Blue-collar	(2) Blue-collar	(3) Blue-collar ^b	(4) Blue-collar	(5) Blue-collar	(6) White-collar	(7) White-collar
Number of observations	709	709	623	709	709	419	419
Method	OLS	OLS	OLS	25th quantile	75th quantile	OLS	OLS
Small	0.172 * * (0.067)	0.181 * * (0.083)	0.158 * * (0.080)	0.094 (0.099)	0.225 * * (0.084)	0.398 * * (0.092)	0.427 * * (0.140)
Medium	0.295 * * (0.046)	0.129 (0.109)	0.065 (0.094)	–0.011 (0.145)	0.152 (0.124)	0.605 * * (0.093)	0.655 * * (0.176)
Large	0.462 * * (0.077)	0.346 * * (0.133)	0.344 * * (0.165)	0.232 (0.198)	0.089 (0.171)	0.737 * * (0.114)	0.750 * * (0.179)
Union * Micro		–0.031 (0.313)	–0.393 * * (0.172)				0.590 (0.599)
Union * Small		0.139 (0.181)	0.124 (0.156)				–0.023 (0.223)
Union * Medium		0.141 * (0.080)	0.037 (0.081)				–0.022 (0.118)
Union * Large		–0.206 (0.137)	–0.105 (0.148)				0.141 (0.194)
IPSS * Micro		0.079 (0.130)	0.044 (0.116)	0.038 (0.175)	0.091 (0.154)		–0.030 (0.139)
IPSS * Small		–0.024 (0.086)	–0.067 (0.087)	0.100 (0.128)	–0.087 (0.112)		–0.038 (0.124)
IPSS * Medium		0.166 (0.107)	0.209 * * (0.087)	0.401 * * (0.152)	0.087 (0.132)		–0.052 (0.161)
IPSS * Large		0.356 * * (0.118)	0.199 (0.139)	0.450 * * (0.220)	0.300 (0.189)		–0.127 (0.145)
Worker quality and working conditions	No	No	Yes	No	No	No	No
R ²	0.177	0.192	0.437			0.424	0.428
Adjusted R ²	0.169	0.174	0.324			0.414	0.407
Pseudo R ²				0.109	0.112		

* Significant at 10% level.

* * Significant at 5% level.

^aRegressions also include years of schooling, potential labor market experience and its square and an indicator of household headship. Numbers in parentheses are heteroskedasticity-consistent standard errors.

^bSample excludes workers in mining and electricity, and workers for whom worker quality or working condition controls are missing.

spline tenure effects are allowed (not shown), union and IPSS effects are again insignificant, and the pattern of tenure effects remains the same as in Table 5.

The lack of significant union effects contrasts with US results,³⁴ but is reasonable in light of the nature of the union movement in Peru, as discussed above. The IPSS results admit the interpretation that minimum wage compliance plays an important role in explaining size effects for blue-collar workers in medium and large establishments, but the results seem more likely to overstate than understate the minimum wage compliance role. Evidence that IPSS effects are stronger at the bottom of the wage distribution (25th quantile regression in column 5) than at the top (75th quantile regression in column 6), which might be expected if IPSS captures true effects of minimum wage compliance, is at best weak; and IPSS effects rise rather than fall with size, as might be expected if threat effects are important.

The most striking result of this section is the robustness of all size differentials among white-collar workers and of the micro-to-small differential among blue-collar workers. Differences in unionization, minimum wage compliance, worker quality and working conditions fail to explain these wage differences. Profit-maximizing rationales like those discussed in Section 4.2 must be important in explaining them. Given the endogeneity of compliance, such rationales may also help explain the size effects associated with minimum wage compliance.

4.2. Efficiency wage considerations

The analysis thus far suggests that some establishments pay premium wages, that either the share of establishments paying premiums or the size of the premiums rises as establishment size rises, and that the payment of premiums is not (for the most part) forced on larger establishments by unions or government wage regulations. Thus, it appears to be in some employers' interest to pay premium wages even in the absence of external constraints. Economic, sociological and political considerations suggest a variety of reasons why this may be so.

Purely economic variants of efficiency wage models identify three sets of concerns that may give some employers reason to pay workers more than they could expect to receive elsewhere. In the most popular model, some employers find it costly to monitor workers' effort directly, and choose to reduce shirking by combining offers of wages above what workers could expect to get elsewhere with the threat of firing workers caught shirking (Bulow and Summers, 1986). Several models point to the use of high wages to reduce hiring, training and downtime

³⁴ US union–nonunion differences are significant, but controlling for them leaves substantial employer size–wage effects unexplained for the US (Brown and Medoff, 1989) and Germany (Schmidt and Zimmermann, 1991). Larger wage effects of industry unionization levels in larger firms suggest some role for unionization threat explanations of the size–wage association, but do not explain the association away (Pearce, 1990).

costs associated with turnover by reducing quit rates (Stiglitz, 1974), reducing the probability that recruits will decline offers (Lang, 1991), or increasing the rate of applications per vacancy (Montgomery, 1991). Finally, costs of screening applicants combined with the need for higher quality workers may cause some employers to pay premium wages in order to draw into the applicant pool more high quality workers, whose reservation wages are higher (Weiss, 1980). These concerns may lead to average wage differentials by establishment size if differences by size in technology, organization or rents lead to differences by size in the share of employers choosing to pursue high wage strategies (rather than monitoring, training or screening directly), or in the size of the optimal premiums.³⁵

Sociological and political considerations lead to two other potential explanations for the employer size–wage effect. High wages might be used to increase individual workers' diligence or the quality of cooperation among workers by improving their 'morale' or 'esprit de corps' (Clague, 1977), or by serving as a gift for which workers are willing to reciprocate with greater effort (Akerlof, 1982).³⁶ A simple political economy story would suggest that larger establishments pay higher wages in order to remain in the government's good graces, and thus to maintain access to rents.³⁷

Although I cannot test the validity of these explanations econometrically, I can narrow the range of explanations most meriting further research. Most important, shirking-based efficiency wage explanations may be ruled out for Peru. In such models, higher wages produce lower shirking only if combined with a credible threat that workers will be dismissed if caught shirking. Peruvian job security legislation, however, greatly reduces the credibility of such threats by making dismissal very costly and difficult, especially in large establishments. In 1985–

³⁵ Higher capital–labor ratios, more sophisticated technology, higher ratios of workers to owners, and deeper hierarchies in larger establishments may increase costs of monitoring, shirking, hiring, training, screening or failing to get workers of high quality. Differences in rents by establishment size may also lead to differences in the incidence or level of efficiency wages. For example, managers may prefer to make their life easier by substituting high wages for direct screening, training and monitoring of workers, and may have more latitude to indulge their preferences in establishments earning rents (Slichter, 1950). Current employees (insiders) may also be able to bargain themselves into a share of larger establishment rents, because employers would incur costs of hiring, training and sabotage (a form of shirking) if insiders were replaced by outsiders at market-clearing wages (Lindbeck and Snower, 1986).

³⁶ Higher monitoring costs, greater importance of cooperation among employees, greater difficulty in using close personal relations between employer and employee to instill good morale, or higher rents (increasing workers' sense of fair wages) may render such efficiency wages more important in larger establishments.

³⁷ Schultz (1982) finds an association between wages and industries' effective protection in Colombia, suggesting that some workers do capture rents bestowed by government on producers. Controlling for industry did not diminish size premiums in this paper; thus, protection differentiated only by sector cannot explain the size effect in Peru. Explanations would have to rest on differential access to benefits like subsidized credit by establishment size within privileged sectors.

1986, workers could be fired freely only in their first 3 months on the job. After 3 months, workers had to be given 3 months' notice or 3 months' wages as indemnity for dismissal. After 3 years on the job they acquired 'absolute stability', in which they could be fired only for specific grave offenses mentioned in the law.³⁸ The burden of proof was on the employer, who also had to follow bureaucratic dismissal procedures. If the employer failed to prove grave offence, he could be required to pay penalties of up to 12 months salary and back pay of up to 6 months salary, and could be required to take the employee back in the same position (Reyes, 1986).

Evidence that the law greatly increased the cost of firing workers, for at least some employers and probably the larger ones, is of various sorts. Historical developments suggest that Peruvian job security legislation was binding on enough employers to make it an important political issue. When first instituted in 1970, absolute security began after just 3 months on the job. In 1978, the perceived costliness of the initial legislation led to modifications limiting absolute stability to workers with more than 3 years on the job (Reyes, 1986; p. 19). Emergency exceptions to the law were also made for sectors in crisis (p. 21). Job security continued to be important in political debates in 1985, and in June 1986 the Garcia administration reextended absolute stability to workers with over 3 months on the job, although this was soon followed by a variety of emergency exceptions. Statistical evidence is more sketchy, but also leads to the conclusion that the legislation was enforced for at least some employers. Reyes (1986) (p. 30) reports regression results showing that the elasticity of employment with respect to production fell significantly between the 1955–1969 period (in which there was no job security legislation) and the 1970s; and Marshall (1993) concludes from a comparative study of job tenure distributions in seven Latin American countries that restrictive regulations on dismissals are associated with lower turnover rates and higher fractions of workers with many years in the firm (p. 12).³⁹

The political economy story seems unlikely to be important also. It would be relevant for explaining wage differences between a few very large firms—who would attract the government's attention—and all others; but is less relevant for explaining why wages in establishments with more than five workers pay large premiums relative to establishments with five or fewer workers. The inability of the IPSS effect to explain much of the wage variation across establishment sizes casts further doubt on the importance of this explanation, because employers concerned with pleasing the government would be likely to comply with labor legislation.

³⁸ In principle, bureaucratic approval could be obtained for layoffs with 'technical or economic' rationales, but in practice approval took years and seldom was obtained (Reyes, 1986; p. 60).

³⁹ Less formal but perhaps more telling evidence is drawn from informal interviews conducted by Jaana Remes in late 1994, in which employers complained about not being able to get 'stable' workers to work hard.

Of the remaining explanations, I find least compelling the model in which larger employers use high wages to improve the average quality of the applicant pool. For this mechanism to explain large wage premiums, workers would have to differ greatly in exogenous qualities that are uncorrelated with standard human capital variables and are unobserved by researchers. More important, these qualities must be unobservable by employers even after workers have been on the job for some time.

Explanations involving larger employers' use of high wages to maintain steadier workforces are much more compelling. Secondary evidence shows that capital–labor ratios rise with establishment size, reflecting likely increases in the cost of vacancies and differences in technology that might require more worker training.⁴⁰ Turnover-related efficiency wage stories also provide easy explanations for a wide range of observations. Lower apparent quit rates in larger establishments, and steeper wage–tenure associations—especially at low tenures—in larger establishments, are consistent with larger employers' greater interest in reducing training costs and other turnover costs. In addition, the more likely importance of firm-specific training among white- than blue-collar workers could explain the larger employer size–wage effect among blue-collar workers; and differences across industries in the relationship between technology and size might explain differences in the employer size–wage effect across industries (not shown). It is also plausible that differences in turnover cost are especially large between micro and small establishments. This size threshold may mark the difference between artisanal and more modern operations, in which the division of labor renders each worker's productivity more dependent on specific knowledge about the organization and on whether other positions are vacant or filled.

Morale-based efficiency wage explanations are a close second to turnover-based explanations. One could probably identify differences in monitoring technology that lead to differences across occupations and industries in larger employers' choices between direct monitoring and the use of good morale to promote high productivity. The use of high wages to maintain morale may also become especially large at a low establishment size threshold, if close personal relations with the employer are a viable substitute for high wages in producing good morale only in micro establishments. While it would be difficult to reconcile morale-based efficiency wage stories with rising wage–tenure profiles for individual workers, such stories are compatible with positive cross-section correlations between tenure and wages; high wages make workers less likely to quit and more likely to be observed at higher tenures. The main conceptual weakness of sociological models is, as Carmichael (1990) puts it, that the theory 'has such a short distance between

⁴⁰ Villaran et al. (1988), for example, estimated capital per workers at US\$600 in enterprises with 1–4 workers (excluding artisan shops), US\$3000 in enterprises with 5–19 workers, and US\$12,000 and US\$40,000 in enterprises with 20–199 and over 200 workers.

its assumptions and its conclusions' (p. 291); but this is a call for more careful theoretical and empirical research rather than a reason to reject the explanation.⁴¹

In sum, shirking-based efficiency wage and simple political economy explanations for the payment of premiums by larger establishments may be ruled out. Of the remaining explanations for the employer size–wage effect considered in this section, turnover- and morale-related efficiency wage explanations are the most compelling.

5. Implications for future research

Having summarized in the introduction the insights the study provides about the operation of labor markets in Peru and similar developing countries, I conclude with two lessons for future research. The first lesson comes from the quantitative result that the largest and most robust size effects, and the size effects that are most different between the US and Peru, are between micro (2–5 employees) and small (6–20 employees) establishments. This suggests that future research on urban production and employment in developing countries should be careful to examine differences by size among micro and small establishments. Studies employing industrial census and survey data, which often pertain only to establishments with at least 10 employees, are likely to miss much of what is important to the employer size–wage effect and much of what differs between developing and developed country labor markets. The second lesson comes from the qualitative results on which explanations for the employer size–wage effect appear most compelling. More detailed empirical and theoretical research on workplace norms that affect wage determination, and especially on recruiting and training practices, would be valuable.

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⁴¹ Sociological considerations may have another role to play if turnover efficiency wage models are important. As Salop (1979) points out, the need to pay workers premiums in turnover models arises out of employers' inability to pay lower wages for new workers than for trained workers. Without such a constraint, employers should be able to push initial wages low enough (possibly negative) that workers become indifferent between these and other jobs despite the expectation of quit-reducing high wages at later tenures (some turnover-related models, like Lang (1991) and Montgomery (1991), are not subject to this critique, but only because they place emphasis on filling vacancies quickly rather than on reducing turnover). Why initial wages are not pushed lower is a puzzle, which sociological considerations may help solve.

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Appendix A. Variable definitions

Variable	Definition
Cash wage	Real hourly cash wage for main job in last seven days in June 1985 Intis. Deflator calculated by Glewwe (1988) pertains to global expenditures in last 3 months and varies by region and date of interview.
Total wage	Real hourly wage including bonuses and cash value of in-kind payments for main job in last 7 days in June 1985 Intis. Deflator same as for cash wage. In-kind payments include those in meals, groceries, housing, clothing and transport. For 26 observations with positive but missing values for one or more in-kind payments, missing pieced were filled in using predicted values from regressions on schooling, experience and its square and industry indicators. Results are little affected by setting these values to zero or omitting these observations.
(Potential) experience	Age minus highest grade attained minus years of school repeated minus five.
Tenure	Months or years (as indicated) worked in main job in last seven days. Tenures of X months plus 2 or more weeks rounded to $X + 1$.
Years schooling	Highest grade attained.
Years primary	Highest grade attained in primary school, ranging from zero to six.
Years secondary	Highest grade attained in secondary school, ranging from zero to five.
Years post-second	Years of post-secondary school, ranging from zero to five.
University degree	Equals one if the highest schooling level attained is university and the individual reports having received a diploma in the highest level attained.

Technical degree	Equals one if the highest schooling level attained is post-secondary non-university and the individual reports having received a diploma in the highest level attained.
Public school	Equals one if the last school attended was public.
Father's schooling	Highest grade attained by the individual's father.
Off-job training	Equals one if the individual reports having taken a training course in an academy, occupational training institute, superior technical institute or military technical institute, or by correspondence.
Off-job certification	Equals one if Off-job training equals one and the individual reports having obtained a diploma or certificate in that course.
On-job training	Equals one if the individual reports having taken a training course in a working place or enterprise.
On-job certification	Equals one if On-job training equals one and the individual reports having obtained a diploma or certificate in that course.
Lima	Equals one if the individual resides in Lima.
Usual weekly hours	Usual weekly hours worked in the main job of the last 7 days.
Boss is relative	Equals one if the individual reports that his employer, manager or director is related to him.
Commute minutes	Individual's report of how long it takes to get to work from home.
Motor transport	Equals one if the individual takes public or private motor vehicle transport to work and zero if the individual walks.
Training Now	Equals one if the individual reports doing a training course at the time of interview.
Union	Equals one if the individual reports that there is a labor union or other kind of workers' organization at his place of work.
IPSS	Equals one if the individual reports receiving the benefits of the Instituto Peruano de Seguridad Social.

References

- Akerlof, G.A., 1982. Labor contracts as partial gift exchange. *Q. J. Economics* 97, 543–569.
- Akerlof, G.A., Yellen, J.L., 1986. Introduction. In: Akerlof, G.A., Yellen, J.L. (Eds.), *Efficiency Wage Models of the Labor Market*, Cambridge Univ. Press, Cambridge, pp. 1–21.
- Aw, B.-Y., Batra, G., 1994. Exports, Firm-Size and Wage Inequalities. Pennsylvania State Univ., processed.

- Barron, J.M., Black, D.A., Loewenstein, M.A., 1987. Employer size: the implications for search, training, capital investment, starting wages and wage growth. *J. Labor Economics* 5, 76–89.
- Becker, G., 1962. Investment in human capital: a theoretical analysis. *J. Political Economy* 70, 9–49.
- Bollinger, W., 1987. Peru. In: Greenfield, G., Maram, S. (Eds.), *Latin American Labor Organizations*, Greenwood Press, New York, pp. 607–666.
- Brown, C., Medoff, J., 1989. The employer size–wage effect. *J. Political Economy* 97, 1027–1059.
- Bulow, J., Summers, L., 1986. A theory of dual labor markets with application to industrial policy, discrimination, and keynesian unemployment. *J. Labor Economics* 4, 376–413.
- Carmichael, H.L., 1990. Efficiency wage models of unemployment—one view. *Economic Inquiry* 28, 269–295.
- Clague, C., 1977. Information costs, corporate hierarchies, and earnings inequality. *Am. Economic Rev. Papers Proc.* 67, 81–85.
- Dickens, W.T., 1986. *Wages, Employment and the Threat of Collective Action by Workers*. National Bureau of Economic Research Working Paper No. 1856.
- Duncan, G.J., Stafford, F.P., 1980. Do union members receive compensating wage differential?. *Am. Economic Rev.* 70, 355–371.
- Evans, D.S., Leighton, L.S., 1989. Why do smaller firms pay less?. *J. Human Resources* 24, 299–318.
- Fajana, O., 1975. Intraindustry wage differentials in Nigeria. *J. Developing Areas* 9, 523–538.
- Gerlach, K., Schmidt, E.M., 1990. Firm size and wages. *Labour Rev. Labour Economics and Industrial Relations* 4, 27–50.
- Glewwe, P., 1988. *The Distribution of Welfare in Peru 1985–1986. Living Standards Measurement Survey Working Paper No. 42*. The World Bank, Washington, DC.
- Goldberger, A., 1983. Abnormal Selection Bias. In: Karlin, S., Amemiya, T., Goodman, L.A. (Eds.), *Studies in Econometrics, Time Series and Multivariate Statistics*. Academic Press, pp. 67–88.
- Gonzales, G., 1980. *Neuvo Derecho Peruano del Trabajo*. Libreria Studium, Lima.
- Griliches, Z., 1979. Sibling models and data in economics: beginnings of a survey. *J. Political Economy* 87, S37–S64.
- Grootaert, C., Arriagada, A.-M., 1986. *The Peruvian Living Standards Survey: An Annotated Questionnaire*. The World Bank, Washington, DC.
- Hashimoto, M., 1981. Firm-specific human capital as a shared investment. *Am. Economic Rev.* 71, 475–482.
- Heckman, J.J., 1979. Sample selection as a specification error. *Econometrica* 46, 153–161.
- Idson, T., 1989. Establishment size differentials in internal mobility. *Rev. Economics and Statistics* 71, 721–724.
- Idson, T., Feaster, D., 1990. A selectivity model of employer–size wage differentials. *J. Labor Economics* 8 (1), 99–122.
- Kruse, D., 1992. Supervision, working conditions, and the employer size–wage effect. *Industrial Relations* 31, 229–249.
- Lang, K., 1991. Persistent wage dispersion and involuntary unemployment. *Q. J. Economics* 106, 181–202.
- Lazear, E., 1981. Why is there mandatory retirement?. *J. Political Economy* 87, 1261–1284.
- Lee, L.-F., 1983. Generalized econometric models with selectivity. *Econometrica* 51, 507–511.
- Lindbeck, A., Snower, D., 1986. Wage Rigidity, Union Activity and Unemployment. In: Beckerman, W. (Ed.), *Wage Rigidity and Unemployment*. Johns Hopkins Univ. Press, Baltimore, MD, pp. 97–125.
- Little, I.M.D., Mazumdar, D., Page, J.M., 1987. *Small Manufacturing Enterprises: A Comparative Study of India and Other Economies*. Oxford Univ. Press, New York.
- Marshall, A., 1993. *Economic Consequences of Labour Protection Regimes: A Comparative Study of Latin American Countries*. Int. Inst. for Labour Studies Discussion Paper 58.
- Mellow, W., 1982. Employer size and wages. *Rev. Economics and Statistics* 64, 495–501.

- Montgomery, J.D., 1991. Equilibrium wage dispersion and interindustry wage differentials. *Q. J. Economics* 106, 163–179.
- Pastor, M., 1992. *Inflation, Stabilization and Debt: Macroeconomic Experiments in Peru and Bolivia*, Westview, Boulder.
- Paus, E.A., 1991. Adjustment and development in Latin America: the failure of Peruvian heterodoxy, 1985–1990. *World Dev.* 19, 411–434.
- Pearce, J.E., 1990. Tenure, unions and the relationship between employer size and wages. *J. Labor Economics* 8, 251–269.
- Reyes, R.E., 1986. *Las Leyes de Estabilidad Laboral y sus Efectos Económicos*. Instituto Libertad y Democracia, Lima.
- Salop, S.C., 1979. A model of the natural rate of unemployment. *Am. Economic Rev.* 69, 117–125.
- Salop, S., Salop, J., 1976. Self-selection and turnover in the labor market. *Q. J. Economics* 91, 619–628.
- Schmidt, C.M., Zimmermann, K.F., 1991. Work characteristics, firm size and wages. *Rev. Economics and Statistics* 73, 705–710.
- Schultz, T.P., 1982. Effective protection and the distribution of personal income by sector in Colombia. In: Krueger, A. (Ed.), *Trade and Employment in Developing Countries*, Vol. 2 Univ. of Chicago Press, Chicago, pp. 83–148.
- Schultz, T.P., 1990. Women's changing participation in the labor force: a world perspective. *Economic Dev. and Cultural Change* 38, 457–488.
- Slichter, S., 1950. Notes on the structure of wages. *Rev. Economics and Statistics* 32, 80–91.
- Smith, R.E., 1987. *Politica salarial, mercado de trabajo y salarios industriales en San Pablo, 1960–1976: analisis segun tamaño de las empresas y su condicion de nacionales o extranjeras*. *Desarrollo Económico* 27, 399–421.
- Stiglitz, J.E., 1974. Alternative theories of wage determination and unemployment in LDCs: the labor turnover model. *Q. J. Economics* 88, 194–227.
- van der Gaag, J., Stelcner, M., Vijverberg, W., 1989. Wage differentials and moonlighting by civil servants: evidence from Côte d'Ivoire and Peru. *World Bank Economic Rev.* 3, 67–95.
- Velenchik, A.D., 1995. *Government Intervention, Efficiency Wages, and the Employer-Size Wage Effect in Zimbabwe*. Wellesley College Working Paper #95-09.
- Villaran, F., Alvarez, S., Mendoza, C., 1988. *Perspectivas del Desarrollo de la Pequeña y Micro Industria en un Contexto de Crisis Económica*, Centro de Ingeniería para el Desarrollo Laboral-CIPDEL, Lima.
- Weiss, A., 1980. Job queues and layoffs in labor markets with flexible wages. *J. Political Economy* 88, 526–538.